

Dissertation

A Distributed Edge FLISR Solution & Network Simulation Test Platform

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Abstract

The energy sector is facing wide-scale changes with the rapid introduction of disruptive distributed energy sources, renewables, electric vehicles and the changing relationship between the consumer and the utility. This has resulted in the need to leverage the capability of modern ICT and IoT technologies to meet the challenges presented by such changes. Of key concern is the resilience of the energy grid and the growing dependency on a stable supply to support business as usual activities and daily life for society at large. This details an attempt to realise a novel implementation of the concept of Fault Identification, Isolation and Service Restoration in a distributed fashion, enabled by Edge Computing techniques, to mitigate the effects of service loss to the end consumer and utility. The research focuses on the Irish energy grid in particular, exploring the background and motivation for such resiliency solutions, the related literature, research activities and the insights obtained from the gathered results.

Declaration

I have read and understood the Departmental policy on plagiarism.

I declare that this document is my own work and has not been submitted in any form for another degree or diploma at any university or other institution of tertiary education.

Information derived from the published or unpublished work of others has been acknowledged in the text and a list of references is given.

Signature _____
Darren Leniston

Date _____

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Glossary

Advanced Distribution Management System (ADMS) A software platform containing a suite of functions to support distribution network management and optimisation.

Customer Minutes Lost (CML) Measurement of the duration of interruption of supply over the course of a year. Consists of the average CML per consumer per year, in the event a supply interruption lasts up to or more than 3 minutes.

Distribution System Operator (DSO) Designated entity responsible for management of the distribution network from source of generation to final consumer.

Fault Passage Indicator (FPI) Indicator notifying that an on-pole switch has detected a fault on its specific point of the network.

Geographical Information System (GIS) Software system utilised to analyse and visualise data containing geographic information.

Load Break Fault Make Switch (LBFM) On-pole circuit breaker which can be controlled to energise or de-energise a section of network in the event of over-load, fault event or required maintenance.

Loss of Voltage Indicator (LVI) Indicator notifying that an on-pole switch has lost supply on its specific point of the network.

Low Voltage (LV) Network Distribution network which transmits to end customers, generally at mains rated voltage (230V single phase & 400V 3 phase for the Irish LV network).

Medium Voltage (MV) Network Distribution network which transmits at voltages above 1kV (kilovolt) (10kV, 20kV & 38kV for the Irish MV network).

List of Acronyms

CI Customer Interruption

DA Distribution Automation

FLISR Fault Location, Isolation and Service Restoration

ICT Information and Communication Technology

IGSD Intelligent Grid Switching Devices

IoT Internet of Things

KPI Key Performance Indicator

MAS Multi Agent Approach

1 Introduction

With the rapid changes as a result of the integration of advanced ICT & IoT technologies into business as usual and daily activities, this has led to large bodies of research in an effort to maximise the potential these technologies and their insights can unlock. Likewise, the energy sector has seen rapid changes with the introduction of renewable energy sources, electric vehicles and a drive to utilise software based solutions which leverage embedded devices. This drive towards the smart grid forms the backdrop of the following dissertation, in particular, this research proposal focuses on the aspect of energy grid resiliency. This dissertation expands on the learning's from previous related research as part of a European Horizon 2020 project [10], to develop a novel approach to an automated grid resilience strategy in the context of the Irish energy grid.

1.1 Research background

The energy grid is of vital importance in today's society and is a key piece of infrastructure which enables activities across a range of industries, services and domestic settings. Like many aspects of the modern world then, the energy industry has been seeing a paradigm shift in the last number of years with the introduction of state-of-the-art ICT and IoT technologies in an effort to modernise the energy sector; in terms of production, transmission and delivery to consumers [5]. These new technologies are taking the form of smart sensors and power monitoring at both the grid and domestic level, unlocking new data and granting greater insight into the state of the grid [2]. This, combined with controllable devices is leading to Advanced Distribution Management Systems (ADMS), enabling a fine-grained view and greater control of the energy grid.

There are a number of factors driving this race to modernise the grid. While traditional energy infrastructures operated on a top-down model, where supply to consumers from the transmission system follows a uni-directional flow and power generation is more centralised [11], this model of energy generation and supply is transforming at a rapid pace. With the introduction of distributed and renewable sources of energy production, electric vehicles, microgrids and the changing relationship between the consumer and the utility, whereby certain consumers are becoming "prosumers" (i.e., generating and selling their excess generation back to the grid) the structure of the energy grid is evolving [6]. In this way, the modernisation of the grid through ICT and IoT technologies intends to meet the challenges posed by the disruptive nature of these changes.

Within this context, the research pursued in this document aims to meet the requirement for a resilient energy grid, utilising modern technologies and techniques, to reduce and mitigate the effects of disruptive events on the grid and their impact on the consumer and utility. To meet this goal, the research leverages the concept of Fault Location, Isolation and Service Restoration (FLISR), a grid resilience strategy which in the event of a fault on the grid, aims to automate the identification of said fault, isolating damaged sections of the network and restoring service to consumers on lines effected by loss of service but are otherwise healthy and undamaged.

In combination with the FLISR technique, the research also explored the application of Edge Computing and novel representations of grid topology data to realise a distributed, fault-tolerant FLISR solution which is not fully reliant on a centralised cloud deployment, by moving subsets of the grid topology and FLISR logic to low-cost edge devices installed on existing grid infrastructure. In order to test, evaluate and demonstrate the solution, a simulation platform was also developed. This research is framed in the context of the Irish distribution grid, in particular three distribution grids of varying topology and features located throughout the country.

1.2 Motivations for the research

In tandem with the rapid transformation of the energy sector, the need for a resilient grid is vital with the growing reliance on consistent power delivery for societal, business and manufacturing activities. Coupled with this is the rising concern of climate change and its impact on everyday life. Recent times have seen an increase in the number and severity of weather events which cause localised and wide-scale damage to infrastructure. Such events often lead to loss of energy supply to consumers, businesses and other institutions through damage caused to wires, transformers, substations, equipment and other grid assets [15]. In relation to the Irish energy network, weather events such as ex-hurricane Ophelia in 2017 resulted in 5,500 damaged overhead lines and 385,000 homes & businesses without power [7].

In addition, the Distribution System Operator (DSO) in Ireland, ESB Networks, is faced with fines by the regulator based on specific KPIs which are directly influenced by the amount of unplanned service interruptions and loss of supply, namely Customer Interruption (CI) & Customer Minutes Lost (CML). For reference, Ophelia resulted in a total of 665.6 million CML, culminating in a 4.86 million euro fine imposed by the Irish energy regulator [7]. However, it must be said that although Ophelia was a rare event, 2017 was not an outlier in terms of total CML per customer, as detailed in ESB Networks 2020 report [13] and illustrated in Figure 1.

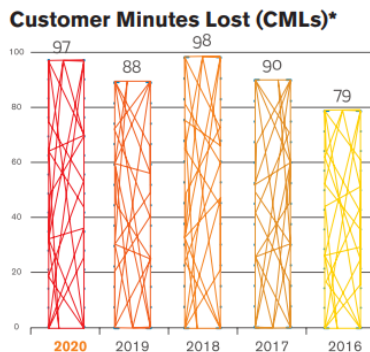


Figure 1: CML figures 2016-2020

In this way then, techniques such as FLISR offer the possibility to drastically mitigate against the effects of inclement weather events. However, there are limitations to the DSO in regards to deploying a FLISR solution to the grid, such as the cost of specialized equipment, installation and requirement for large-scale software platforms such as ADMS. In combination with this, weather events can also effect communication networks, thereby reducing or even negating the benefit of a cloud-based or centralised FLISR implementation. With this factor in mind, the concept of edge computing is one that may greatly benefit the operation of a FLISR algorithm in the field, by moving a subset of the business logic and data processing away from a centralised system, to the edge of the network. In the context of FLISR, this would enable sections of the network to operate independently if needed to isolate and restore fault events, providing a fault-tolerant FLISR solution able to operate effectively in conditions such as Ophelia.

1.3 Research objectives

The key objectives which framed the direction of the research activities centered around delivering a novel FLISR solution, leveraging modern technologies which can operate both centrally and in a distributed fashion at the network edge, and to test, evaluate and demonstrate the operation and effectiveness of the solution through the use of a simulation platform, utilising a number of grid topologies to assess the FLISR algorithms effectiveness against

a variety of network configurations located throughout Ireland.

The main output of the research is gauging how the developed FLISR solution may reduce CML in real-world operation, to that end, the following questions guided the research activities.

- **R1 - How effective will the outputs of the FLISR solution be in terms of reducing DSO fault KPIs?**

This question is crucial in regards to evaluating the value of the research activities and FLISR solution implemented, both compared to alternative approaches to the FLISR concept undertaken in other research, and in relation to the tested grid topologies and extrapolated to the Irish energy grid as a whole.

- **R2 - Given a similar set of conditions, how will the results produced by the centralised FLISR algorithm compare to the distributed FLISR algorithms operating locally at the edge of the network?**

A key feature of the distributed FLISR solution is to be in itself fault-tolerant, and therefore able to operate and produce results locally in the event communications with the central, “master” FLISR algorithm is unavailable. Given the limited view of the network granted to the FLISR algorithms operating locally at the edge, in comparison to the overall system awareness of the central FLISR algorithm, will the results produced by both be similar and, crucially, valid.

- **R3 - Given a diverse set of network topologies, will the FLISR solution output valid results for a variety of grid configurations?**

For the FLISR solution to be effective, a key requirement will be to operate and produce valid results given a wide set of grid topologies with varying network configurations in terms of structure, number of nodes and device types.

2 Review of the literature

The following section will explore and review a number of items of literature related to the research topics. The findings of this review are collated by a number of sub-themes; digitalisation of the grid, FLISR and edge computing, which directly relate to the concepts and technologies applied to the distributed edge FLISR solution.

2.1 Digitalisation of the grid

There is a wide range of literature available on the concept of the smart grid, and what this digitalisation and modernisation of the energy network means for the energy industry. *Fang et al.* [5] describe a smart grid as utilising two-way flows of both electricity and data in an aim to create an automated and distributed advanced energy delivery network, and through the use of modern ICT lead to a more efficient and flexible grid which can respond to a variety of events. The article in question describes the requirements for the smart grid from the perspective of three major systems; *smart infrastructure systems*, *smart management systems* and *smart protection systems*. Of most relevance here, are smart infrastructure systems, which are described as supporting two-way data flows via wired and wireless communication methods, and smart protection systems of which failure diagnosis and grid self-healing are a key factor.

An article by *Güngör et al.* [9] discuss the trends resulting in the need for a new, smarter energy infrastructure, taking the US as an example, reporting demand and consumption of electricity has increased by 2.5% annually over 20 years. This growing demand, in combination with factors such as greenhouse emissions, the environmental

impact of extracting fossil-fuels for energy production and their growing scarcity, is driving the need for renewable sources and more flexibility in how the energy grid operates by enabling prosumer activities, microgrids, virtual power plants and increased grid resiliency to meet these challenges.

In the paper, the smart grid is described as a “modern electric power grid infrastructure for improved efficiency, reliability and safety, with smooth integration of renewable and alternative energy sources, through automated control and modern communication technologies”. In addition, they describe how the reliability of the energy infrastructure and supply can be supported and maintained in the smart grid, and how the impact of equipment failure, capacity constraints and natural accidents which lead to supply interruption and outages, can be mitigated by online power system monitoring, diagnostics and protection.

As described in the referenced articles, enabling the smart grid requires reliable information and data flow from a variety of smart devices both in the field and in the domestic setting, therefore the communication methods utilised are of key concern. In a separate article, *Güngör et al.* [8] discuss how wireless communication methods can meet some of the challenges presented by the move to the smart grid. In this paper, it is explained that while traditionally power system monitoring was realised through wired communication via cabling from pole to pole, this method is expensive and therefore not widely implemented today due to its financial impact.

Priya & Malhotra [16] describe how the need for efficient, reliable communication on the smart grid can be achieved by leveraging the capabilities of the telecommunication network, specifically referencing 5G and its fast, low-latency features and how these can meet the challenge of capturing large amounts of data from the field and enable quick, automated control services to manage, balance and ensure the resilience of the grid.

2.2 Fault Location, Isolation & Service Restoration

With resiliency of the network a key concern to the DSO and frequently mentioned as a requirement for the smart grid in the literature, FLISR is one such technique which has been explored and discussed in various research. *Agüero* [1] describes FLISR as a self-healing technique, though does make mention of certain authors preferring the term “self-restoration”, as in reality no system can currently fix physical damage to the grid without direct intervention by DSO personnel. The paper discusses how Distribution Automation (DA) technologies can enable remote monitoring, coordination and operation of distributed network assets at substation, feeder and customer level. Most relevant, is the description of utilising remote, on-pole switching devices as a method of sectionalising and restoring supply to customers in the event of a fault. The paper describes how the need for manual investigation and operating of switching devices on the part of DSO operation staff can be reduced or rendered redundant through automated FLISR, and the benefits of such a solution being the improvement of reliability statistics and the savings made due to the reduced cost of outages.

However, the text posits that leveraging such devices for self-healing is more appropriate for urban and suburban areas, rather than remote rural locations, where radial distribution is more commonplace (i.e., one source of power for consumers on the network). Though it could be argued that resilience strategies such as FLISR are more necessary for rural areas, due to the increased prevalence of faults as a result of deterioration and damage in remote locations.

A report produced by the US Department of Energy in 2014 [14] aimed to gather metrics on five separate FLISR solutions implemented throughout the United States over the course of one year. The findings included a large reduction in the number of customers affected by service loss (270,000) with 38 million fewer CML in total, translating to a reduction of 45% in the amount of interruptions and 51% in CML per fault event. The report specifies that an essential component for successful FLISR operations is the communications networks used to monitor and enable control over switching systems and devices, in combination with suitable interfaces which allow interoperability between various DSO systems and device types on the grid.

Each FLISR system explored utilises various device types, communication methods, modes of operation and location of the deployed system as part of its implementation. For example, a utility named CenterPoint trialed their FLISR system, named “Self-Healing Grid” which is supported by Intelligent Grid Switching Devices (IGSD) and requires manual validation by a System Operator in order to enact control operations. Conversely, a utility named Duke leveraged electronic reclosers, circuit breakers and line sensors for its FLISR system, dubbed “Self-Healing Teams” to enable a fully-automated solution.

The report specifies a number of learning’s gathered by the utilities from the FLISR trials, such as the need for the telecommunications network to be more resilient than the power delivery systems which rely upon them, and the necessity for the utility to consider the additional steps required to successfully deploy and operate a FLISR system. Steps identified included the testing and installation of new equipment, and the frequent firmware and software updates necessary to ensure the system continues to operate effectively due to its autonomous nature, though one utility, Georgia Power, found that developing a simulator to test and validate FLISR eliminated the need for extensive field trials and served as a training tool for operation staff.

A paper by *Eriksson et al.* [4] further discuss the limitations and considerations when implementing FLISR solutions, specifically centralised deployments, such as having a single point of failure and the requirement for the handling of large amounts of real-time data, large network topologies and the processing necessary to ensure consistent operation. They propose a distributed FLISR utilising a Multi Agent Approach (MAS) devices, tested against a radial network commonly found in remote areas, and applying Prim’s Algorithm to devise the FLISR logic. The solution proposed was tested against a cyber-physical testbed, with the results finding that the MAS approach to FLISR provided efficient grid stabilisation and fault restoration.

Most relevant is a report produced by ESB Networks as an output of the H2020 project SOGNO [12]. This project focused on delivering turn-key grid services for power quality monitoring, estimating network state, load prediction and forecasting as well as FLISR. The FLISR implementation described in the report leveraged communication from Load Break Fault Make (LBFM) on-pole circuit breakers in the field. In the event of a fault, these switches communicate a Fault Passage Indicator (FPI) and Loss of Voltage Indicator and in this way it is possible to identify a fault occurrence. ESB Networks notes that a number of their networks currently rely upon manual intervention on the part of control room operation staff to manually identify and resolve fault events when they occur. This need for human intervention greatly increases the time taken to restore supply to consumers, leading to higher CML per event. While the FLISR solution tested as part of the SOGNO project produced good results, ESB Networks noted that its reliance on a centralised instance and telecommunications to operate is a severe limitation.

2.3 Edge computing

Over the last number of years, cloud computing has provided a method for large-scale data collection and processing for multiple use-cases and industries, however this dynamic is changing with the proliferation of IoT.

Shi et al. [17] describe this as the post-cloud era, where embedded IoT will not only serve as a data source but also consume and act on this data at the edge. As more data sources come online, the level of processing required by cloud computing services to collect, parse and act on this data increases. This, coupled with network bandwidth constraints when considering the large amounts of data gathered from distributed sources means new solutions are necessary and the paper proposes the edge computing concept as a solution to these challenges. The literature defines edge computing as enabling technologies which allow computing activities to be performed at the edge of the network on behalf of cloud-based services, therefore working as a data consumer as well as a data producer. One such example is the smartphone, which can serve as both a producer of data for cloud-based services and as a processing device at the edge. An example use-case being health-related applications, where user metrics are

collected and processed on the device, and resulting health data is passed and stored on a cloud-based service.

Deploying and managing edge resources is a key factor when considering edge computing solutions, *Taleb et al.* [18] describe the importance of orchestration frameworks to enable integration and management of edge devices in terms of resource allocation and service management in regards to software and firmware updates. The paper describes how utilising virtualisation, such as containerised software services, supports in packaging, delivering, deploying and updating software services operating on edge computing devices located in the field.

In relation to the smart grid, a paper by *Zhang et al.* [19] proposes that a number of challenges facing the smart grid, such as high computational cost and transmission delay can be met through the application of edge computing. Similarly, *Chen et al.* [3] discuss the benefits of edge computing in supporting smart grid operations, with specific reference to grid resiliency and self-healing, describing how the dependency on centralised and cloud-based systems may lead to less efficient operation of grid control and monitoring services. Furthermore, the proximity of edge computing devices to grid assets may reduce latency when intelligent decisions are made locally, and provide finer-grained location awareness and monitoring of the network for the DSO, this is particularly important at the MV and LV levels where visibility of the grid from a system perspective is greatly reduced.

3 Methods

Methods section will describe the context of the research and the actual work done throughout the dissertation.

3.1 Data & Setup of Experiment Environment

This section will describe the underlying data and work of setting up the experiment environment from the simulation and FLISR Edge perspective.

3.1.1 Trial grid topology features

Describe the trial sites selected to use for simulations and testing, describe the features (network structure, no. of switches etc.)

3.1.2 FLISR Edge solution & algorithm

Describe the FLISR Edge solution, how it is structured, orchestrated, the edge devices, algorithms etc.

3.2 Experiments

This section will describe in detail the actual experiments undertaken with the simulation platform and FLISR algorithms.

3.2.1 Network simulation test platform

Describe simulation platform and its structure, purpose, how it operates etc.

3.2.2 Fault simulation operation

Describe how the FLISR Edge solution and simulation platform interact and operate, how it generates results and how these are presented.

4 Results of the research

Section describing the results gathered from the testing, including the findings, relevance of the results to the DSO and consumer, and future work to expand the solution further.

4.1 Research findings

Describe the findings, how effective the solution is (or not), compare and contrast across the selected trial grids, centralised vs. distributed, limitations etc.,

4.2 Relevance to the DSO

Describe the relevance of the results to the DSO and the consumer, from a grid stability, financial and climate perspective (weather events).

4.3 Future work

5 Conclusion

6 Exploitation Activities

Describe any exploitation activities (papers/presentations etc.)

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